Neelay Ranjan

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Education

Purdue University, College of Engineering — Indianapolis, IN
B.S. in Artificial Intelligence · Concentration: Intelligent Control & Systems · Minor: Mathematics — *Expected May 2027*

- GPA: 3.68 / 4.0 · John Martinson Honors College · Dean's List · CITI: Biomedical Research, RCR, GCP

Research & Work Experience

NASA Ames Research Center — May 2026 – Present
Aeronautics Directorate - Code AF — *Moffett Field / Mountain View, CA*

GenAI Applied Research Intern, SHIFT — *Aug 2026 – Present*

- Fine-tuned Qwen3-Omni-30B with QLoRA for air traffic control (ATC) speech-to-text on ∼1.5 h of real audio (∼900 utterances), reaching **17% WER vs. 20% for the existing Whisper ASR** and 30% zero-shot in 5-fold CV.
- Built a generate-and-rank parser for the en route ontology. A text model and rules propose 16 to 32 candidate values per field, and Kev, a typed decision model, picks one. It **raised value-match accuracy from 40% to 80%** over the best rules-based and LLM attempts, at ∼90% lower latency and ∼400× lower cost than Qwen3 27B.
- Building a text-to-speech synthetic ATC corpus that expands the dataset ∼20× (5% real, 95% synthetic). Its outputs feed a maneuver database matched to historical and live FAA airspace records for safety research.

GenAI Applied Research Intern, SLAAC — *May – Aug 2026*

- Built hazard-aware flight rerouting with diffusion models learned from FAA radar tracks. Guidance applied mid-sampling steers around special-use airspace at inference with no retraining, and **99% of reroutes clear the 25 nm hazard buffer at a median +10 nm (1.1%) added distance**, within 1.2% of the geometric optimum.
- Designed a flight-plan LLM with a custom tokenizer that synthesizes a full day of **∼44,000 FAA-managed flights** matched to historical traffic density, producing simulation-ready traffic days for the software NASA and the FAA use to evaluate future air traffic management (ATM) strategies.

Regenstrief Institute — Feb 2024 – Present
Applied Vision-AI Researcher — *Indianapolis, IN*

- Showed an x0 flipped-objective diffusion segmenter outperforms SAM (ViT-B) on benchmark angiograms in ultra-low-data regimes, reaching **88.2% Dice with just 16 labels**. Its pre-labels bootstrap surgeon labeling and **cut correction time by 75%**.
- Built a modular SAM-x0 tiled refinement pipeline (SAM prior + partial-resolution x0 diffusion refiner) for real-world, region-agnostic angiograms at native 1024 resolution. It reaches **70% Dice with 22 labeled images**, beating both SAM (46%) and raw x0 (32%) at a fraction of full-resolution compute.
- Extended the pipeline to cerebral moyamoya MRA, segmenting vasculature from **only 7 labeled images** with patient-level cross-validation and tiling augmentation, at ∼55% Dice vs. SAM's ∼20%.

Selected Publications

- N. Ranjan, S. Clarke, et al. "Generate-and-rank extraction of air traffic control clearances into an en route ontology." *NASA Ames. Extended abstract for AIAA AVIATION 2027, in preparation.*
- N. Ranjan et al. "Bootstrapping surgeon labeling campaigns with x0-diffusion: label-efficient vessel segmentation of catheter-based angiograms." *First author, in preparation.*
- Co-author, "Helical Antenna for Electromagnetic Field Stimulation in Alzheimer's Disease Therapy." *Presented at IEEE MWSCAS 2026. Led the PCB design team.*

Projects & Applied Research

Solar-Powered Energy-Based Chess Engine — 2025

- Trained an energy-based model that scores and ranks moves from a learned energy landscape over game states, reaching **∼1900–2200 estimated Elo** against a ladder of rated Stockfish depths.
- Embedded the EBM on a Raspberry Pi Zero 2 W (∼1 s per move) as a standalone chess engine **powered solely by onboard solar & supercapacitors**. Quantization tests showed FP32 beat INT8 on ARM, which had no native INT8 kernels.

LLM Design Team Lead, V2X Aircraft Maintenance Assistant (Raytheon divestiture) — Aug 2024 – May 2025

- Led design and fine-tuning of a maintenance-assistant LLM whose two-stage RAG pipeline picks sources from their summaries, then pinpoints and quotes the relevant section, **cutting hallucinations from ∼40% to ∼5%**.

Technical Skills

Generative modeling: diffusion (x0 / v-prediction), energy-based models, foundation-model adaptation (SAM), transformers, LLM fine-tuning & RAG, safety-critical AI, interpretability, multimodal pipelines
ML engineering: Python, C/C++, CUDA, PyTorch, transfer learning, medical image segmentation, quantization, Docker, Slurm / H100, W&B, Linux / cluster administration, Git
Hardware / edge: PCB design, ultra-low-power inference, embedded ML